$Q_1 = 25\%$ Quintile $= \frac{p}{100}(N+1)$

$Q_2 = 50\%$

$Q_3 = 75\%$

$IQR = Q_3 - Q_1$

$LB = Q_1 - 1.5(IQR)$

$UB = Q_3 + 1.5(IQR)$

Outlier Detection Technique.

Type's of Probability Distribution.

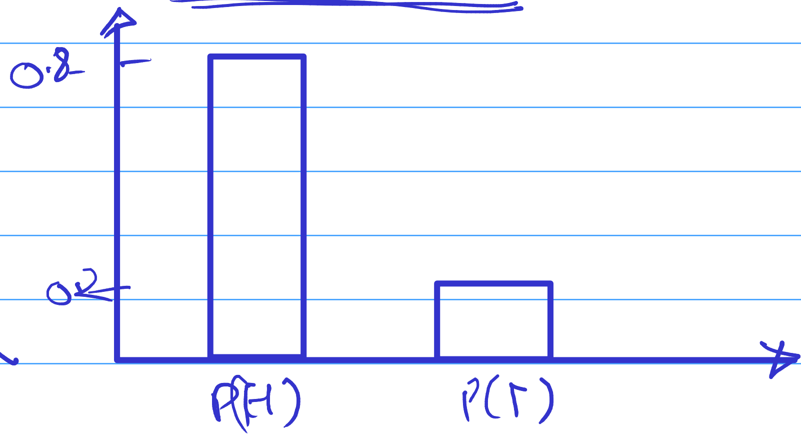
(1) Bernoulli Distribution

- It's Concerned with Discrete Random Variable. whole no
- ⇒ PMF :- Probability Mass Function.
- ⇒ It applies to event that have one trial and two possible outcome.

Total prob = 1

$P(H) = 0.8$

$P(T) = 0.2$



exam → Fail / pass

Survived = 0 / 1

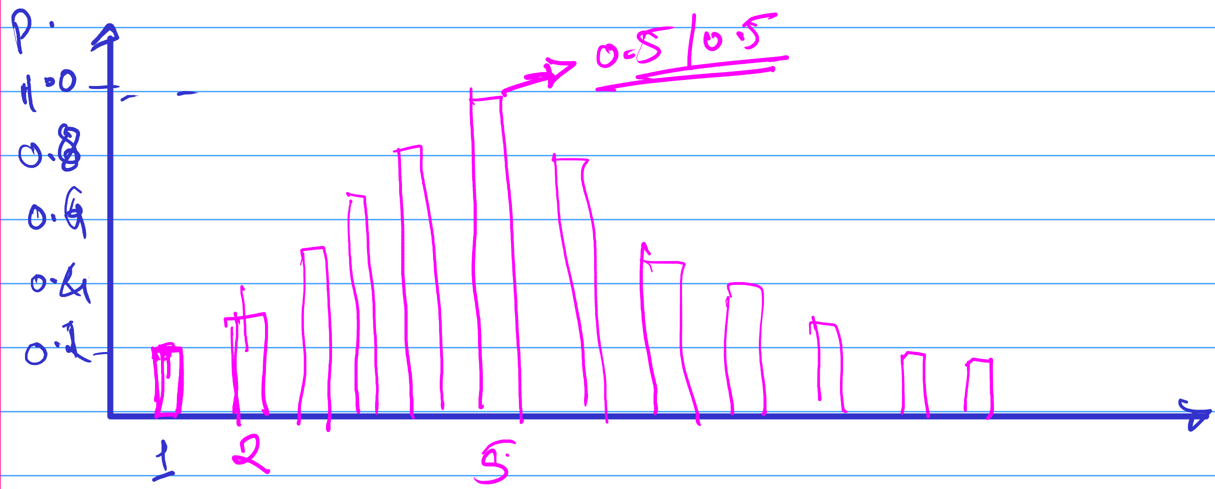
② Binomial Distribution

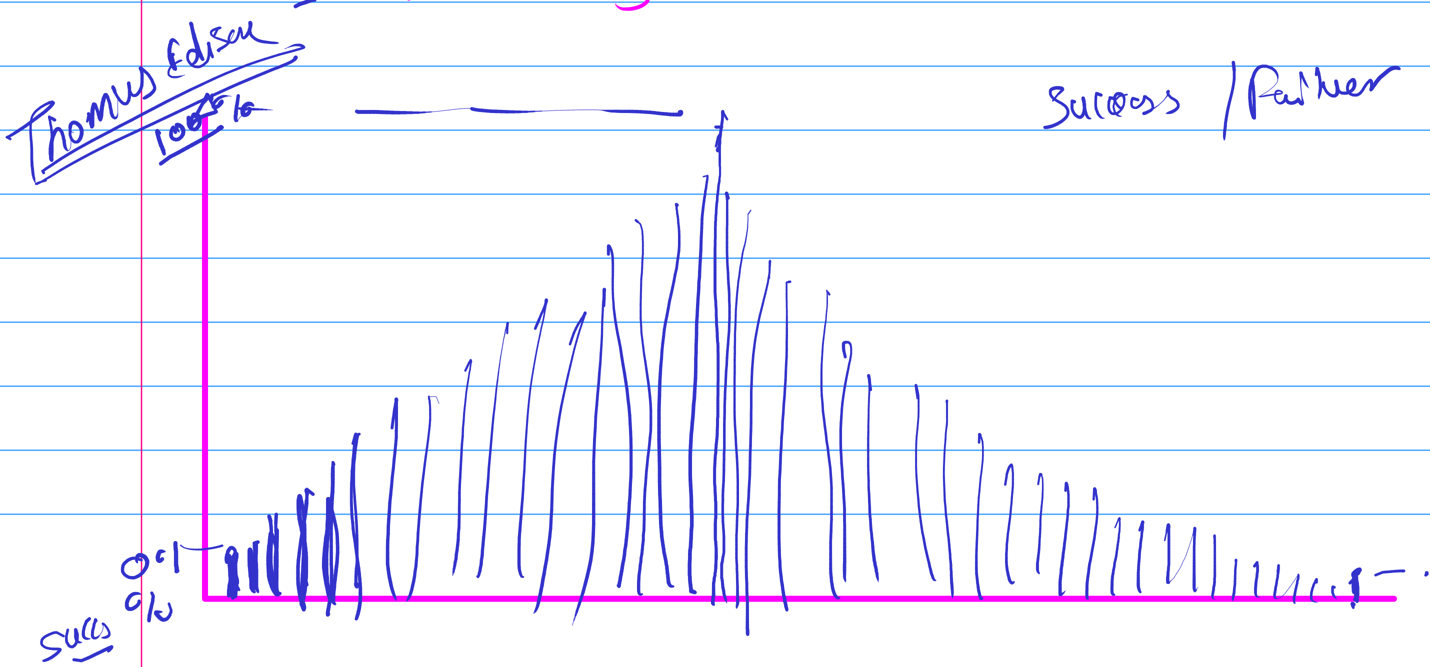
:→ It Concerned with Discrete RV.

:- It follows PMF

:→ There are two possible outcomes but n no. of trials.

Ex = Tossing a Coin 10 times.





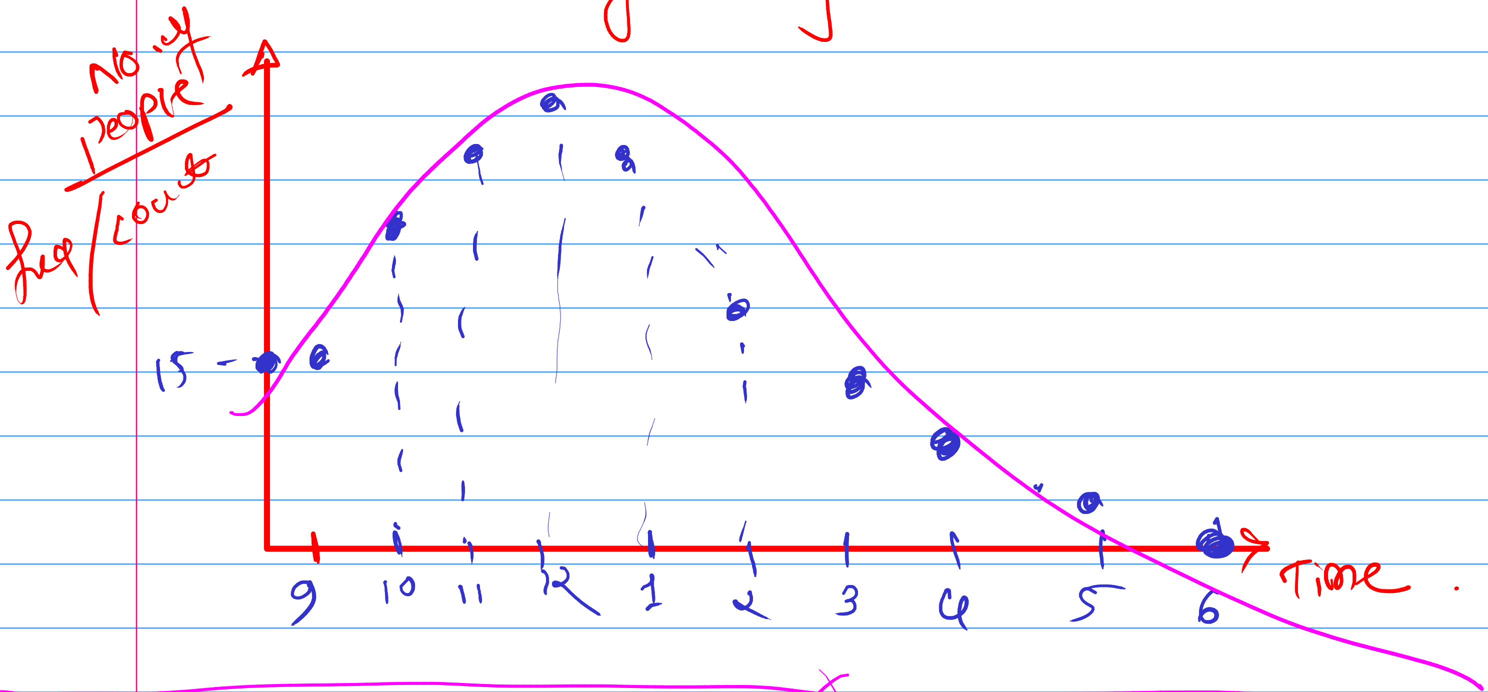
③ Poisson Distribution

:→ PMF

:→ Discrete RV

:⇒ It discribe the no. of events occuring in a fixed time interval.

:⇒ no. of people/customers visit at bank after every Hours.



④ Normal Distribution / Gaussian Distribution

:⇒ CRV → Continious RV → height, weight, BP, kg.

:⇒ Normal Distribution is a symmetrical Distribution.

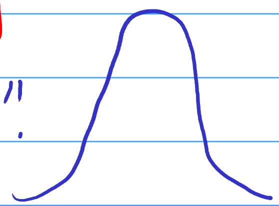
सारखा (Same)

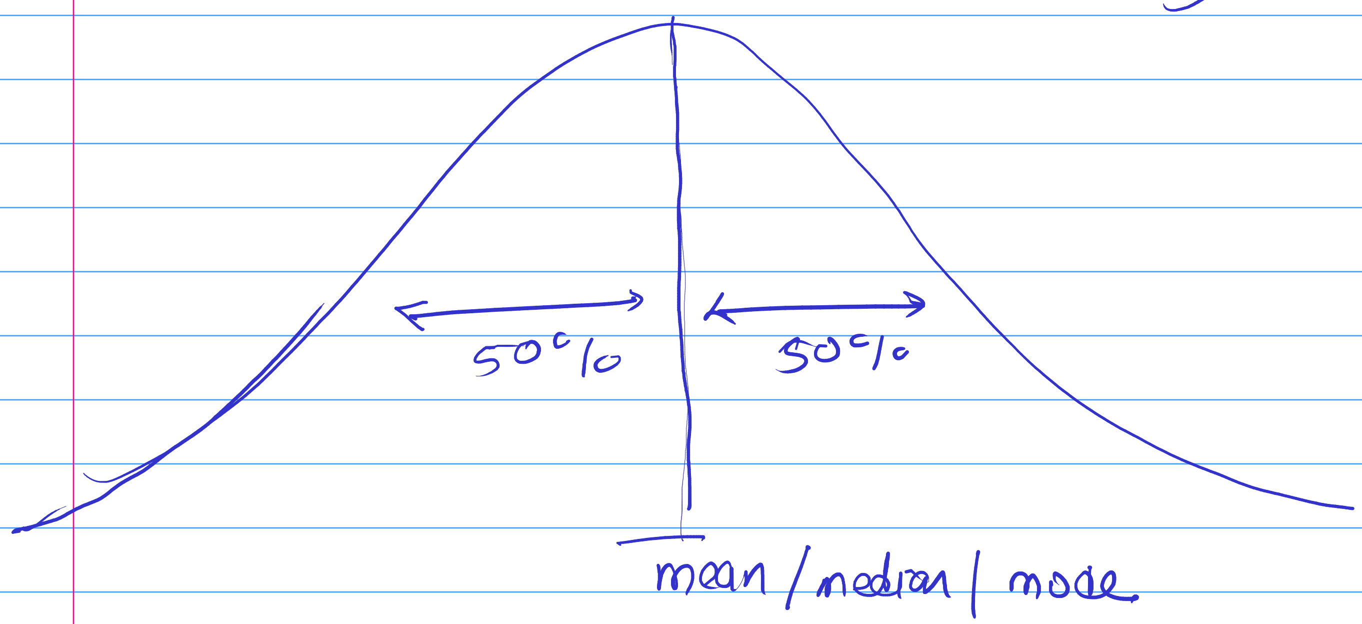
: Sometimes mean = median = mode, when the distribution is not skewed.

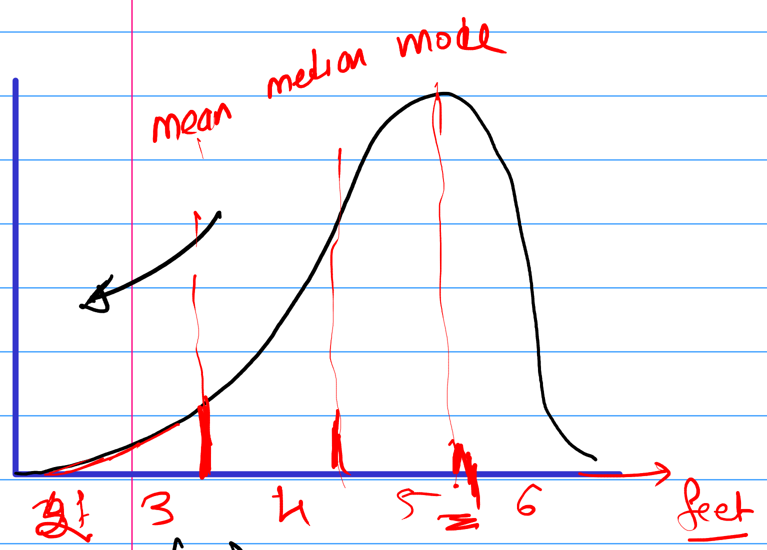
झुकलेला.

→ 50% of data towards left of the mean & 50% of the data towards right.

→ Its look like a "Bell curve".

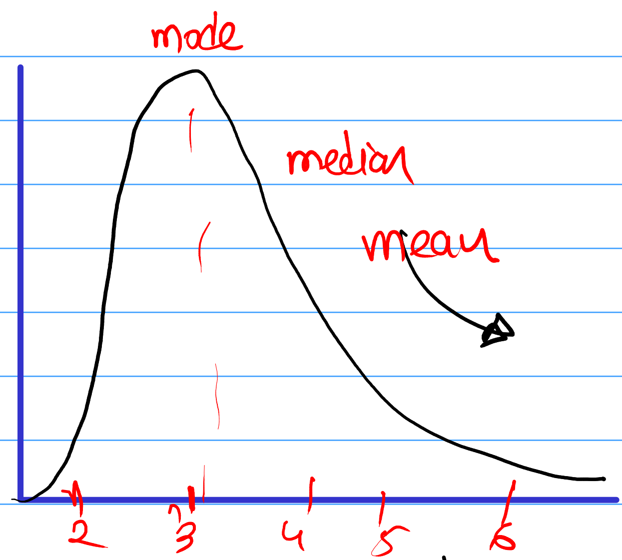






Left Skew

mean < median < mode



Right Skew.

mode < median < mean

Empirical Rule of Normal Distribution

8 → The empirical Rule in statistics also known as 68, 95 & 99 rule.

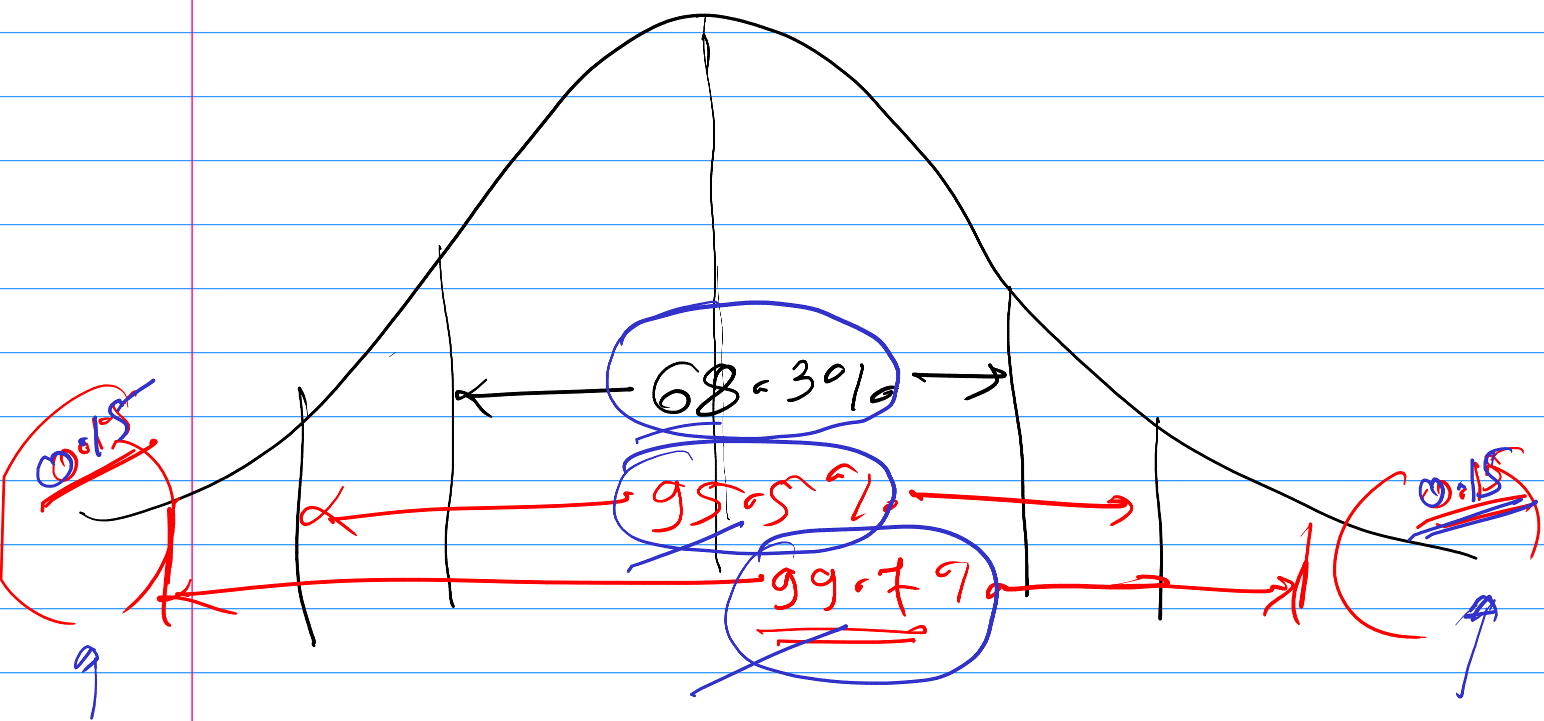
It states that for normal Distributions,

68.3% data lies inside first standard deviation.

95.9% data lies inside Second standard deviation

99.7% data lies inside Third standard deviation.

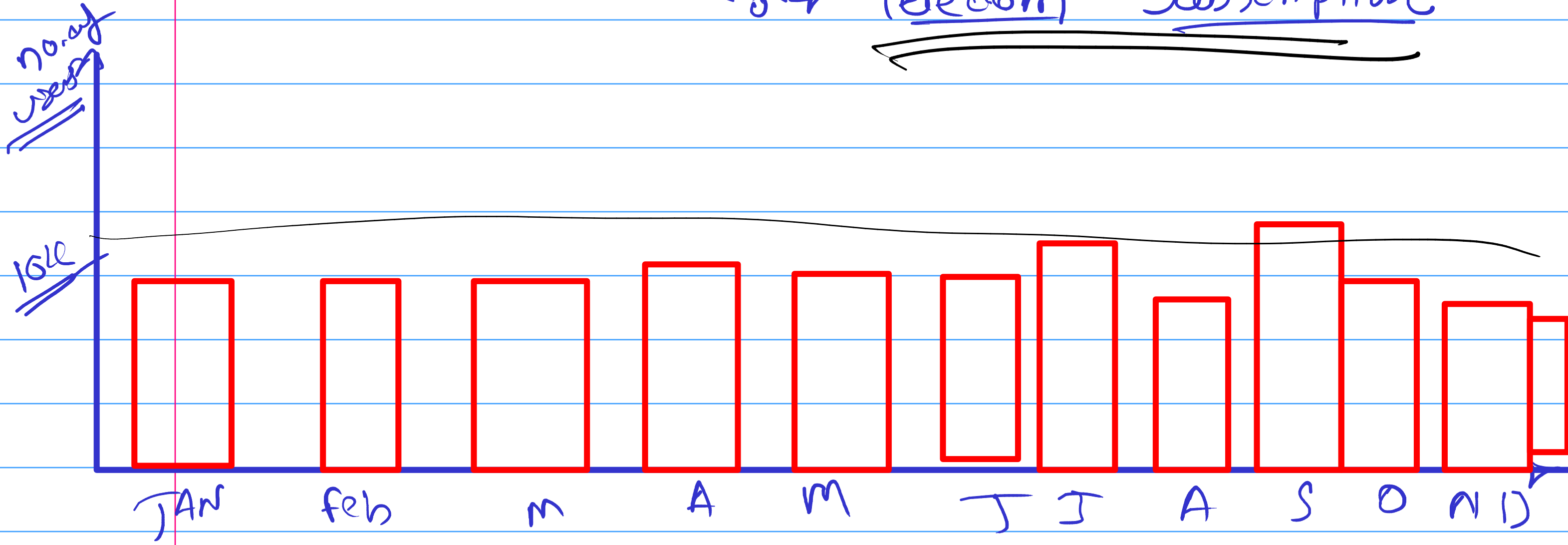
0.3

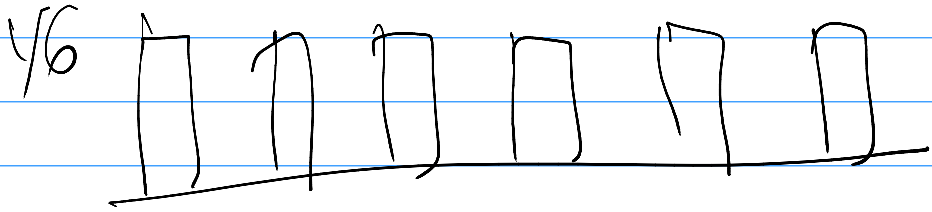


(5) Uniform Distribution

Jandt Kaa

eg ⇒ Telecom Subscription





(6) Log Normal Distribution

$= \log_{10}(\text{data})$

$= \log_{2}(\text{data})$

eg ⇒ ex = Monthly Income of Indian